

Quantifying life history

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Learning objectives

1. Convert life history information into a Leslie matrix
2. Do sensitivity and elasticity analyses

review of last lecture: life tables and structured populations. Now going to do some math on the life history information and calculate population growth rates from life table information

Leslie matrix

- type of population projection matrix
- goal: calculate population growth rate and project abundance in each age
- input: survival and fecundity rates from each age

let's take the stage-structured life table from last lecture and make it a biennial

now ages instead of stages

Age	Description	Number Alive	Survival to Next Stage	Fecundity
0	Seed	1000	0.20	0
1	Seedling	200	0.25	0
2	Adult	50	—	100

let's put this in the leslie matrix format

- fecundity in first row
- each age is a column
- survival of each age on the diagonal

```
leslie <- c(0,0.2,0,0,0,0.25,100,0,0)
matrix(leslie,nrow=3,ncol=3)
```

```
##      [,1] [,2] [,3]
## [1,]  0.0 0.00 100
## [2,]  0.2 0.00  0
## [3,]  0.0 0.25  0
```

can think of 0s as percentage of one stage going to others

for example, [3,1] is 0 because no seeds are going straight to adults

[2,2] is 0 because no seedlings are going back to seeds

Calculating λ

use linear algebra, not going to get into that, just some definitions

- eigenvalue = a scalar used in matrix multiplication
- several eigenvalues for a given matrix *i.e. several scalars can work when doing this transformation/matrix multiplication*
- largest, positive eigenvalue of a Leslie matrix *dominant - largest, positive* = λ = population growth rate

really easy to do in R

Population projection

we can also project the abundances of each age in a given time step

- multiply: Leslie matrix, matrix of N at t
- product: matrix of N at t+1

quick refresher of matrix math

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} ax + by + cz \\ dx + ey + fz \\ gx + hy + iz \end{pmatrix}$$

```
leslie <- c(0,0.2,0,0,0,0.25,100,0,0)
Nt <- c(1000, 200, 50)

leslie.mat <- matrix(leslie,nrow=3,ncol=3)
leslie.mat
```

```
##      [,1] [,2] [,3]
## [1,]  0.0 0.00 100
## [2,]  0.2 0.00  0
## [3,]  0.0 0.25  0
```

```
N.mat <- matrix(Nt,nrow=3,ncol=1)
N.mat
```

```
##      [,1]
## [1,] 1000
## [2,]  200
## [3,]   50
```

```
leslie.mat %*% N.mat
```

```
##      [,1]
## [1,] 5000
## [2,]  200
## [3,]   50
```

Example of stage-structured population

- each column = stage *same as Leslie matrix*
- column 1 = rates from seeds to seeds/seedlings/adults
- column 2 = rates from seedlings to seeds/seedlings/adults
- adults to seeds is fecundity

```
stage <- c(0,0.2,0,0,0.1,0.3,50,0,0.8)
matrix(stage,nrow=3,ncol=3)
```

```
##      [,1] [,2] [,3]
## [1,]  0.0  0.0 50.0
## [2,]  0.2  0.1  0.0
## [3,]  0.0  0.3  0.8
```

Elasticity analyses

let's say you have rare plant population data. population growth rate is less than 1, declining, but you don't know why. Is it survival or reproduction issue?

- quantify influence of vital rates on λ
- proportional change in growth rate

not absolute change in population growth rate, use sensitivity analysis for that

Sensitivity analyses

let's say you could actually improve the survival of the population. Sensitivity analysis tells you how much the population growth rate would increase if you increased survival rate

- stage/age specific increase survival of adults, juveniles, etc
- repeatedly compute population growth rate